SCREENSHOT REPORT

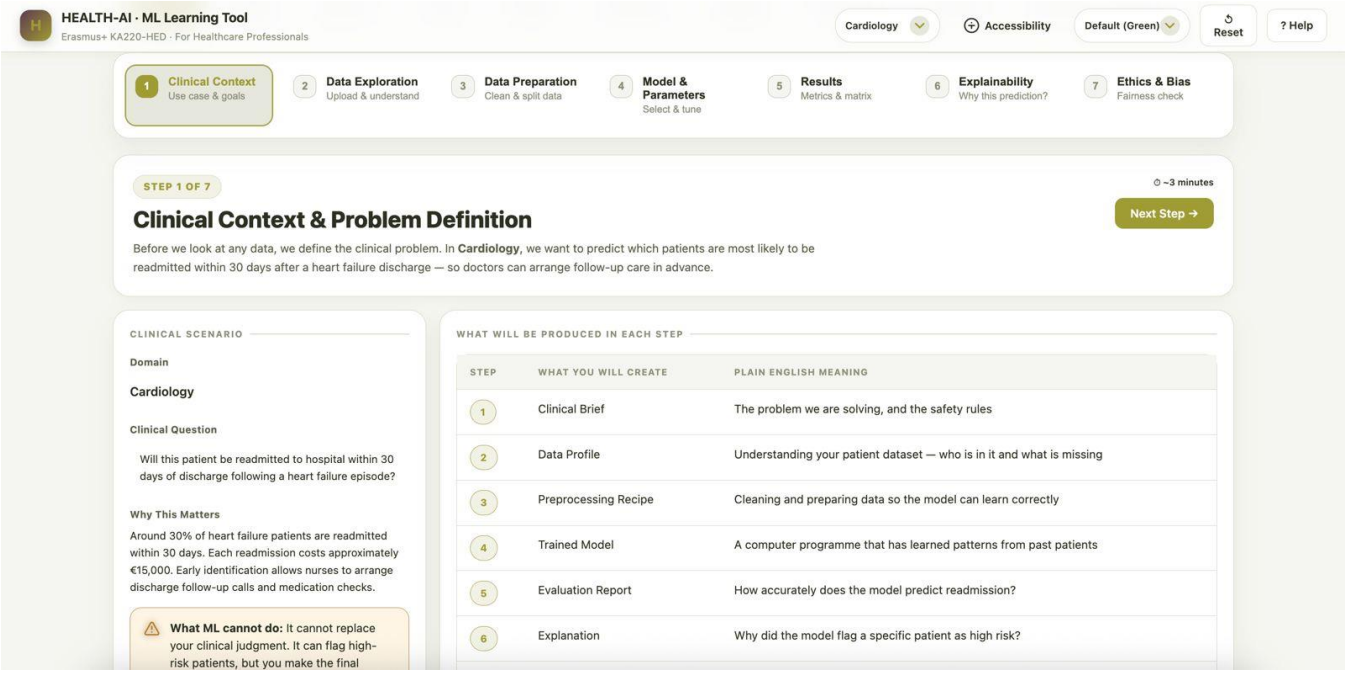
Sprint 2/ Project Documentation

Group Name	Junior Engineers
Group Number	[#00]
Members	Göksu Bulut | Ceren Gökdere | Elif İstanbulluoğlu | Kerim Efe Kocacık | Emre Tek
Date	17.03.2026
GitHub Repo	[https://github.com/...]
Live Demo URL	[https://...]

Prepared as part of course project documentation.

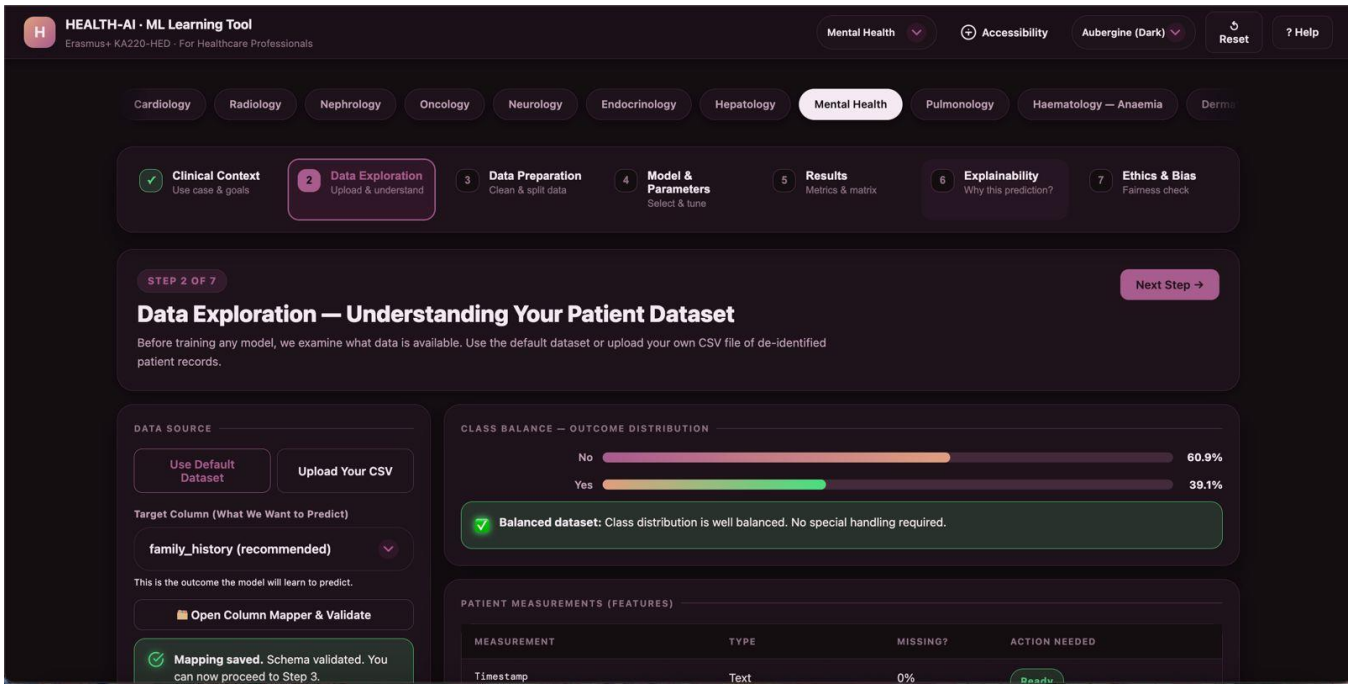
Global Interface Features

Feature 1.1 – Theme Selection (Green Theme)



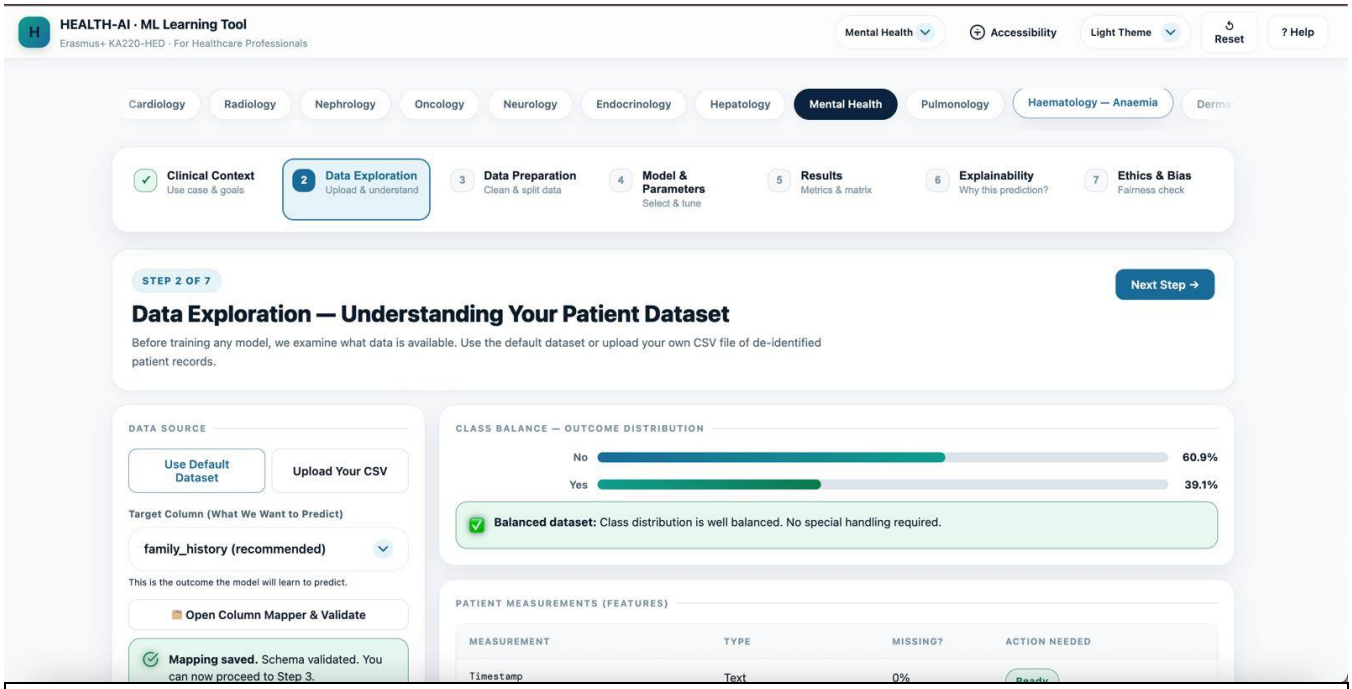
Explanation: This screenshot shows the global interface with the Green theme applied. It demonstrates that users can change the visual theme of the ML Learning Tool through the theme selector in the top navigation bar, improving usability and personalization across the entire application.

Feature 1.2 – Theme Selection (Dark Theme/Aubergine)



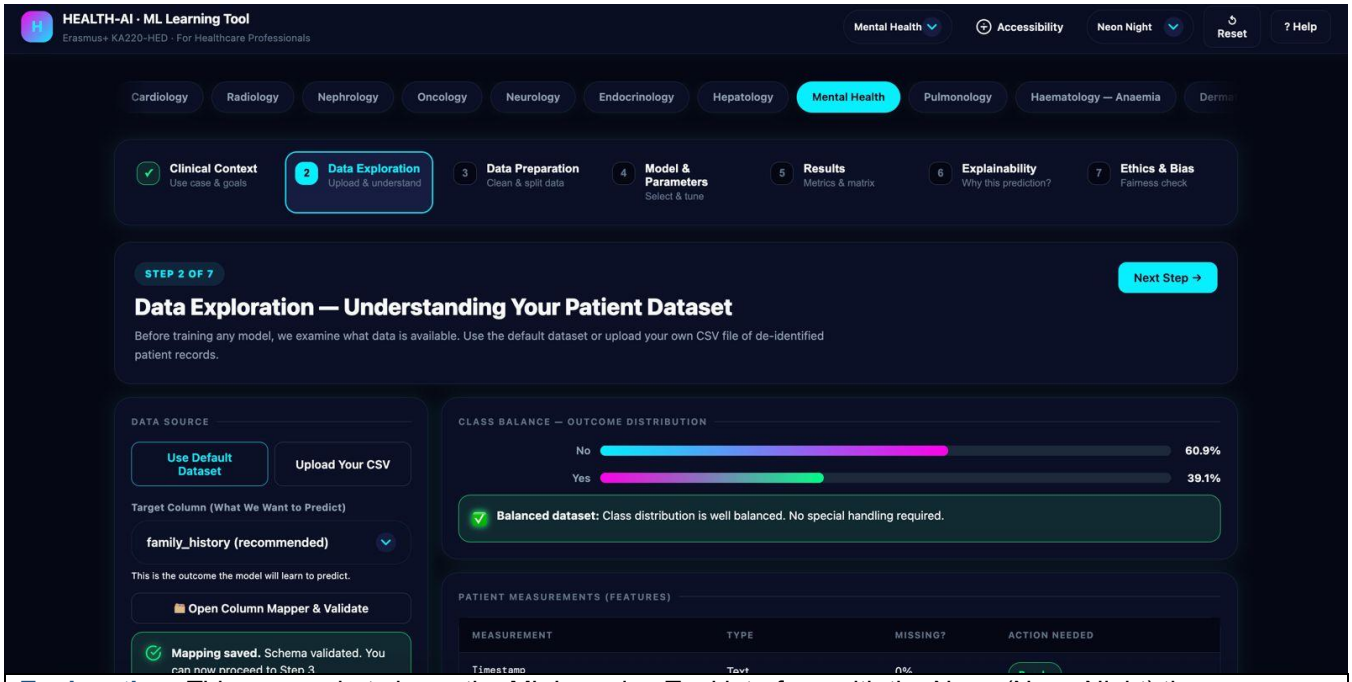
Explanation: This screenshot shows the ML Learning Tool interface with the Dark (Aubergine) theme applied. It demonstrates that users can switch between different visual themes through the theme selector in the top navigation bar, improving accessibility and user preference across the entire application.

Feature 1.3 – Theme Selection (Light Theme)



Explanation: This screenshot shows the ML Learning Tool interface with the Light theme applied. It demonstrates that users can switch between different visual themes using the theme selector in the top navigation bar, allowing the interface to adapt to different visual preferences and improving overall usability.

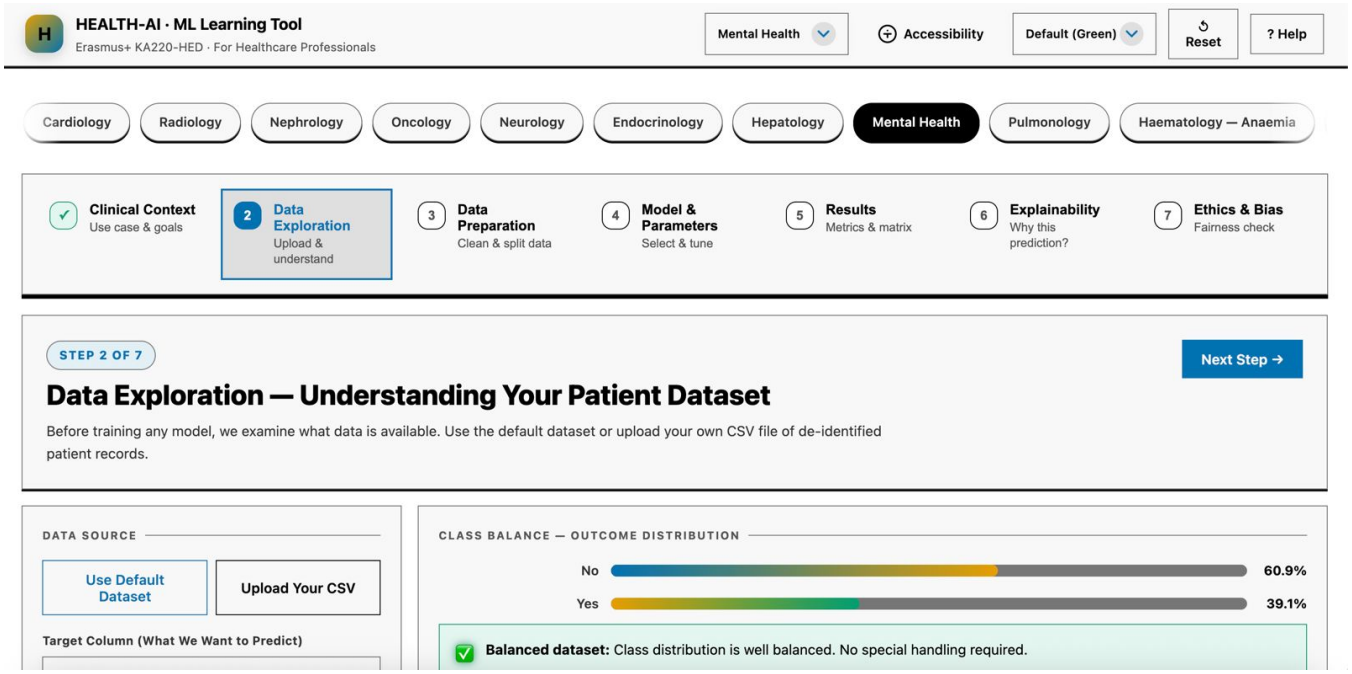
Feature 1.4 – Theme Selection (Neon / Neon Night Theme)



Explanation: This screenshot shows the ML Learning Tool interface with the Neon (Neon Night) theme applied. It demonstrates that users can switch to a high-contrast visual style using the theme selector in the top

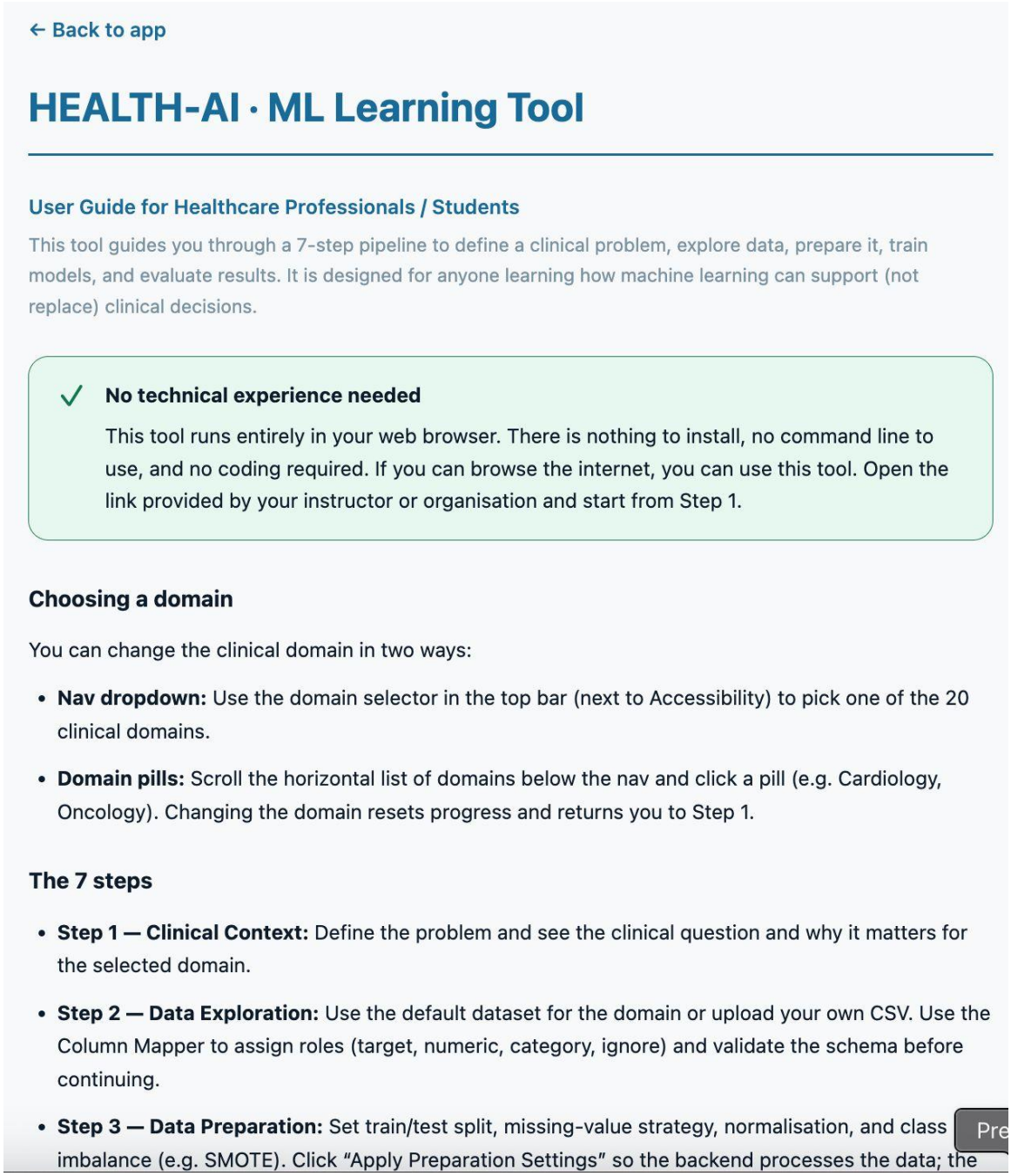
navigation bar, enhancing visual accessibility and providing an alternative interface appearance across the application.

Feature 1.5 – Accessibility Settings (Color-blind Mode and Text Scaling)



Explanation: This screenshot shows the accessibility settings available in the ML Learning Tool interface. It demonstrates that users can enable color-blind friendly visualization and adjust text size, improving readability and ensuring the interface is accessible to users with different visual needs.

Feature 1.6 – Help / User Guide Interface



← Back to app

HEALTH-AI · ML Learning Tool

User Guide for Healthcare Professionals / Students

This tool guides you through a 7-step pipeline to define a clinical problem, explore data, prepare it, train models, and evaluate results. It is designed for anyone learning how machine learning can support (not replace) clinical decisions.

✓ **No technical experience needed**

This tool runs entirely in your web browser. There is nothing to install, no command line to use, and no coding required. If you can browse the internet, you can use this tool. Open the link provided by your instructor or organisation and start from Step 1.

Choosing a domain

You can change the clinical domain in two ways:

- **Nav dropdown:** Use the domain selector in the top bar (next to Accessibility) to pick one of the 20 clinical domains.
- **Domain pills:** Scroll the horizontal list of domains below the nav and click a pill (e.g. Cardiology, Oncology). Changing the domain resets progress and returns you to Step 1.

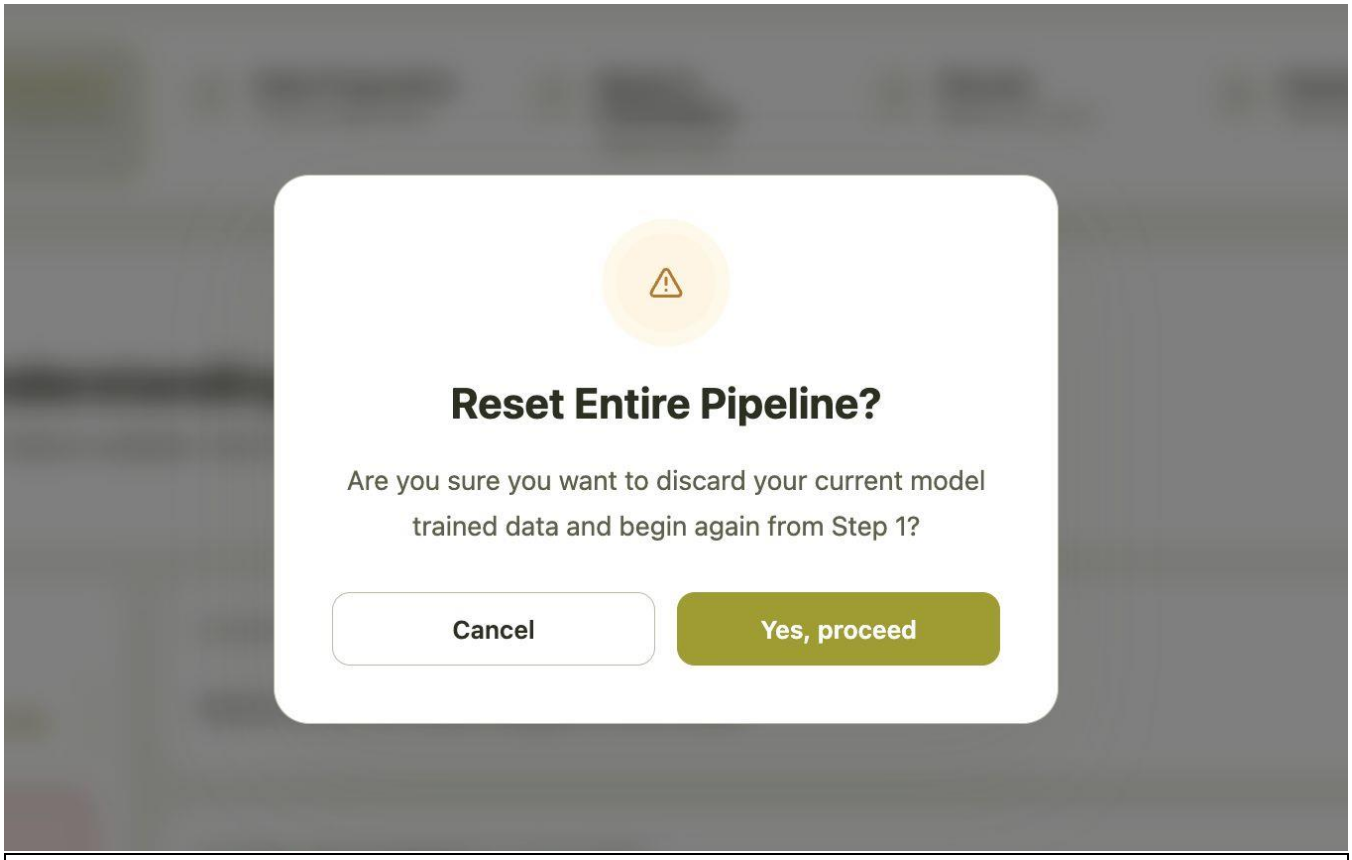
The 7 steps

- **Step 1 — Clinical Context:** Define the problem and see the clinical question and why it matters for the selected domain.
- **Step 2 — Data Exploration:** Use the default dataset for the domain or upload your own CSV. Use the Column Mapper to assign roles (target, numeric, category, ignore) and validate the schema before continuing.
- **Step 3 — Data Preparation:** Set train/test split, missing-value strategy, normalisation, and class imbalance (e.g. SMOTE). Click "Apply Preparation Settings" so the backend processes the data; the

Pre

Explanation: This screenshot shows the Help page that opens when the user clicks the "Help" button in the top navigation bar. It provides a user guide explaining the purpose of the ML Learning Tool, how to select a clinical domain, and an overview of the 7-step machine learning workflow.

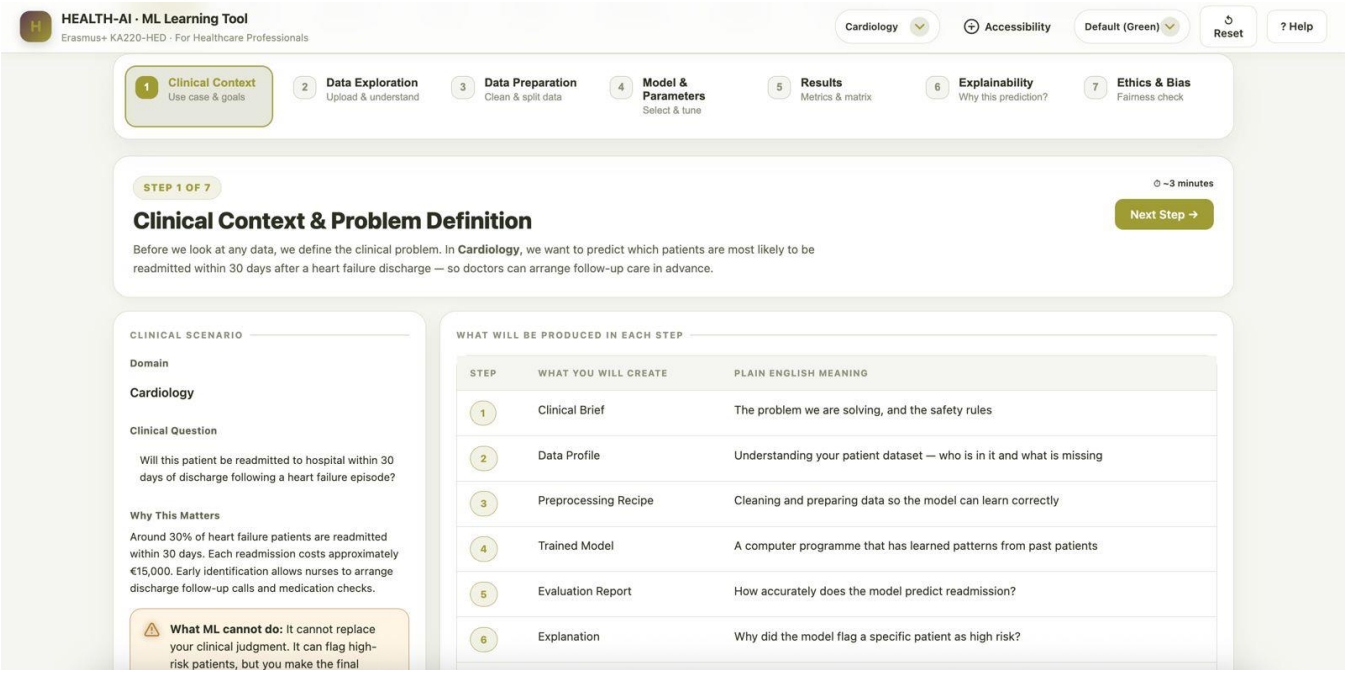
Feature 1.7 – Reset Pipeline Confirmation



Explanation: This screenshot shows the confirmation dialog that appears when the user clicks the “Reset” button in the top navigation bar. It prevents accidental data loss by asking the user to confirm before resetting the entire ML pipeline and returning to Step 1.

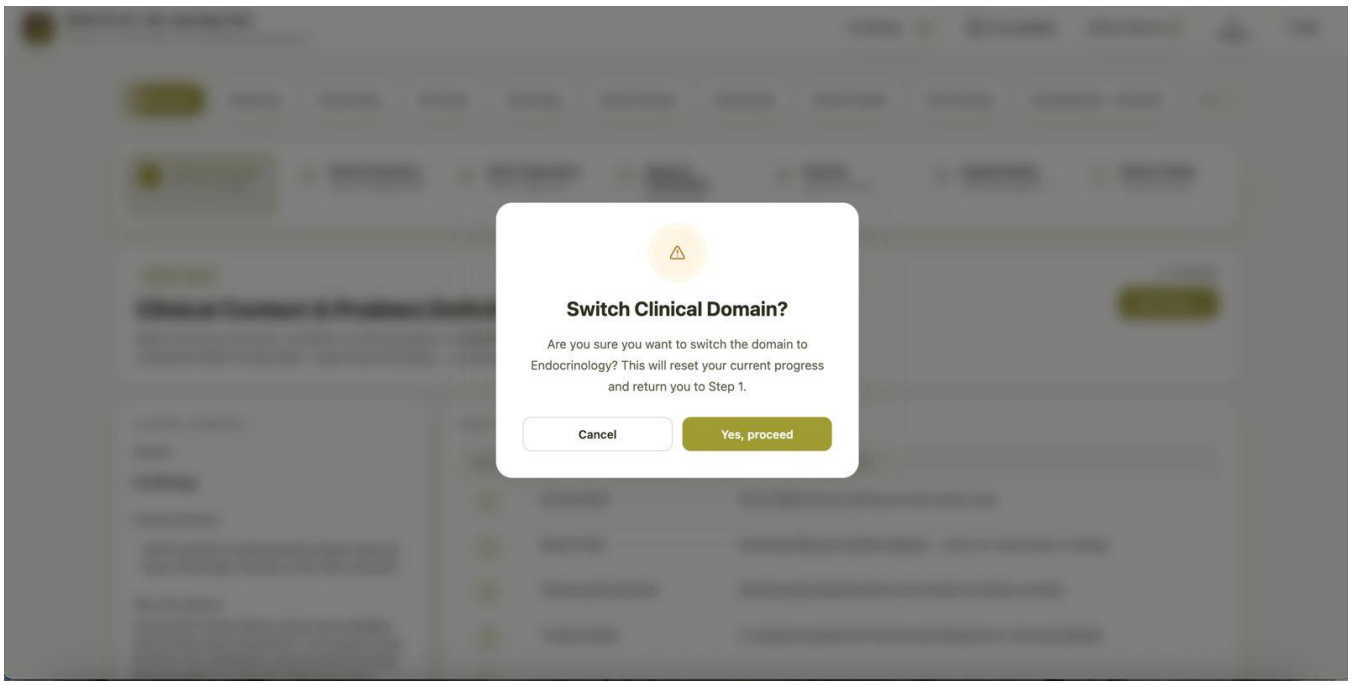
Step 1 – *Clinical Context*

Feature 1.1 – Clinical Context & Problem Definition Interface



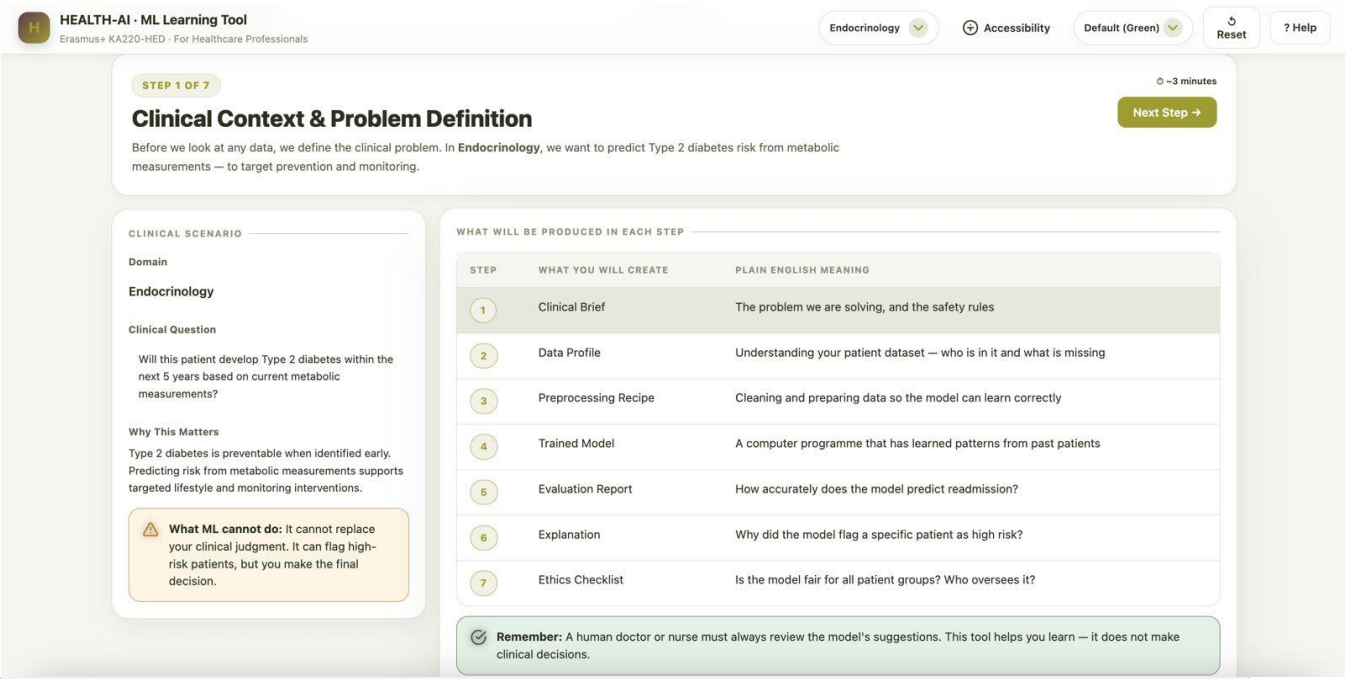
Explanation: This screenshot shows the Step 1 interface where the clinical context and problem definition are introduced. It presents the selected medical domain, the clinical question being addressed, and an overview of the machine learning pipeline to help users understand the problem before exploring the dataset.

Feature 1.2 – Clinical Domain Switch Confirmation



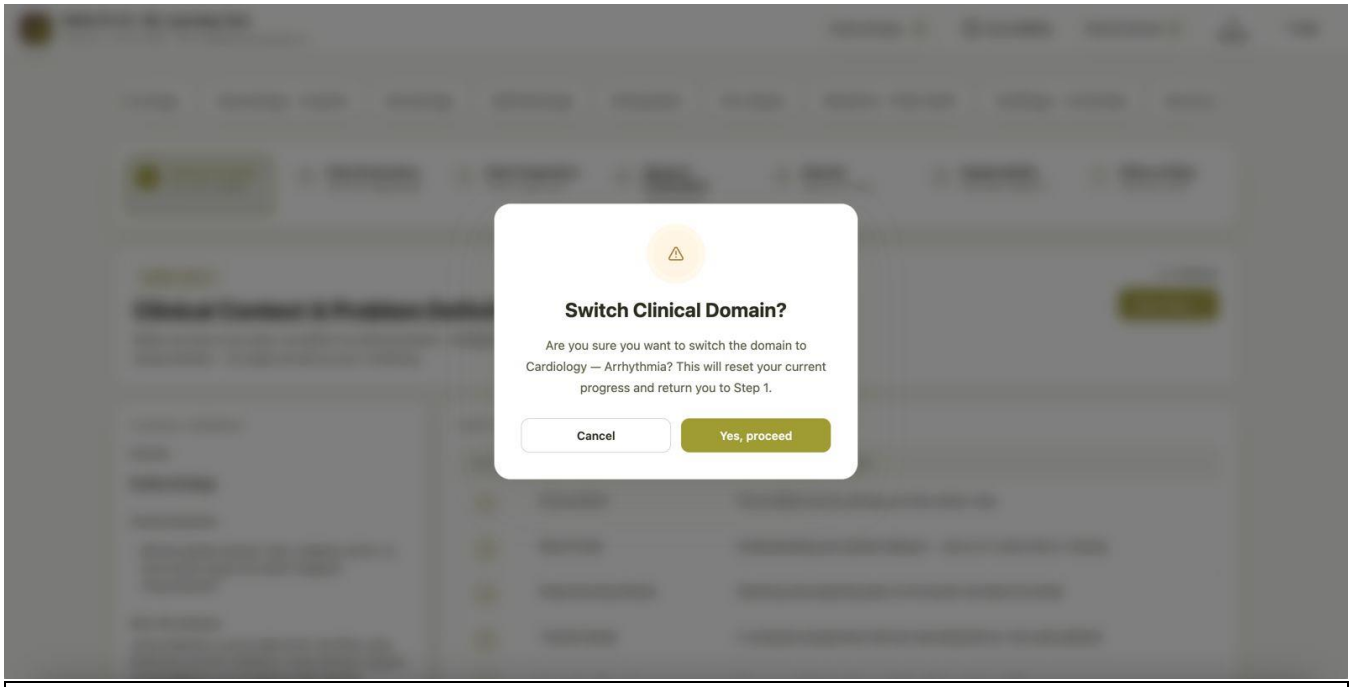
Explanation: This screenshot shows the confirmation dialog that appears when the user attempts to switch the clinical domain. It prevents accidental progress loss by asking the user to confirm the action, since changing the domain resets the current workflow and returns the user to Step 1.

Feature 1.3 – Clinical Context Update After Domain Selection



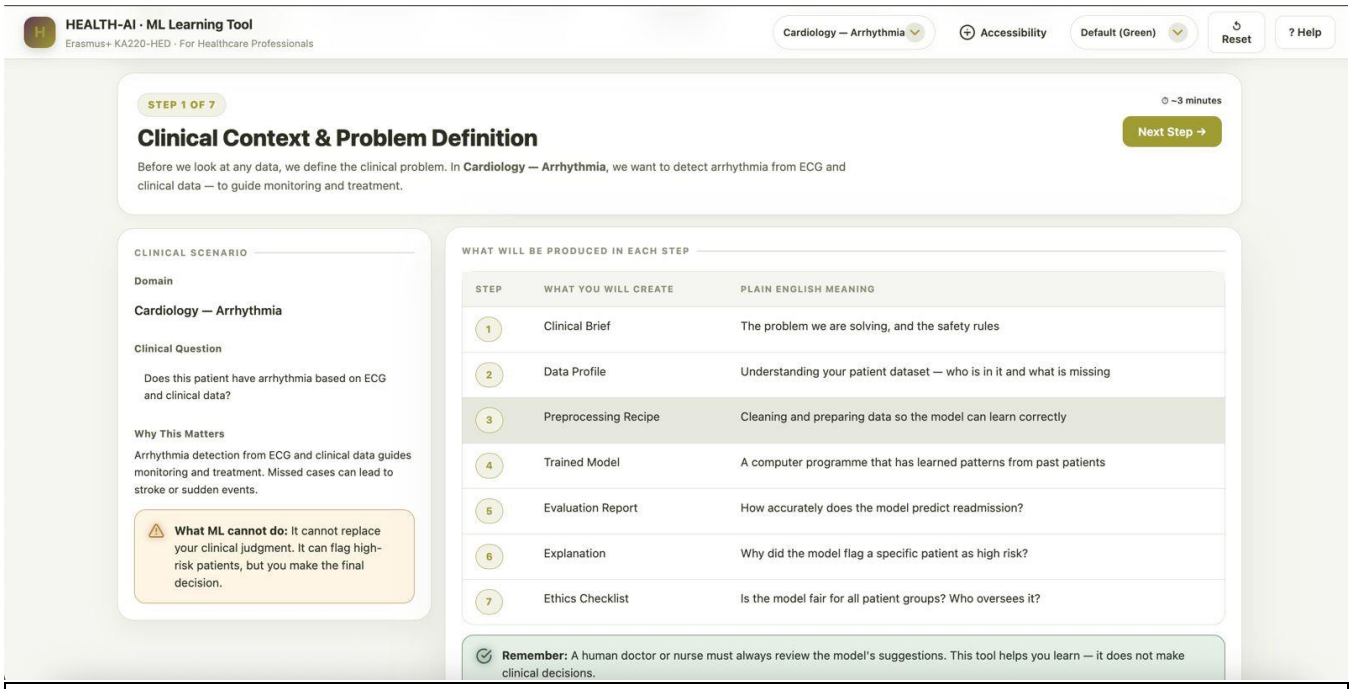
Explanation: This screenshot shows the Step 1 interface after the user confirms switching to a different clinical domain. The clinical context, domain name, and problem description are automatically updated to reflect the newly selected domain (Endocrinology), demonstrating that the system dynamically adapts the workflow to the selected clinical scenario.

Feature 1.4 – Clinical Domain Switch Confirmation (from 2 to 3)



Explanation: This screenshot shows another example of the confirmation dialog that appears when switching between clinical domains. The dialog ensures that users confirm the action before the system resets the current workflow and returns to Step 1.

Feature 1.5 – Clinical Context Adaptation Across Multiple Domains

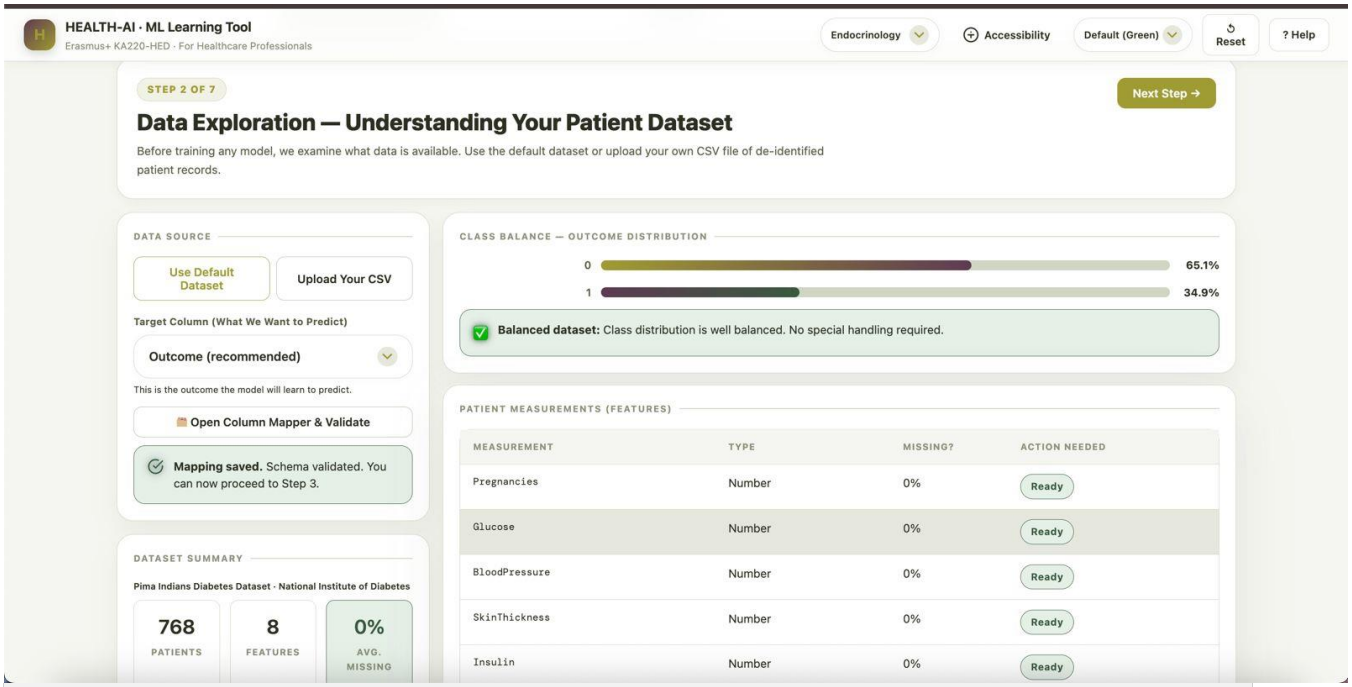


Explanation: This screenshot shows the Clinical Context screen after switching to a third clinical domain (Cardiology — Arrhythmia). The clinical question, domain description, and problem context automatically

update according to the selected domain. This demonstrates that the system supports multiple clinical domains and dynamically adapts the ML workflow context based on the chosen medical specialty.

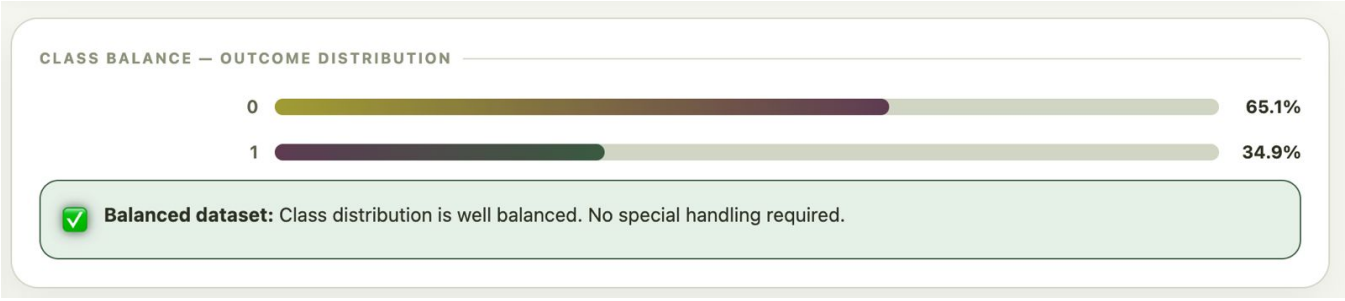
Step 2 – *Data Exploration*

Feature 2.1 – Built-in Example Dataset Loaded



Explanation: This screenshot shows the Data Exploration interface with the built-in example dataset successfully loaded. The system automatically displays the dataset summary, class distribution, and available patient measurements, allowing users to understand the structure and balance of the dataset before proceeding to data preparation.

Feature 2.2 – Class Distribution Analysis



Explanation: This screenshot shows the class balance analysis for the selected dataset. The system automatically calculates the distribution of outcome classes and visualizes the proportion of each

class. In this example, the dataset is identified as balanced, and the tool informs the user that no additional handling (such as resampling) is required before training the model.

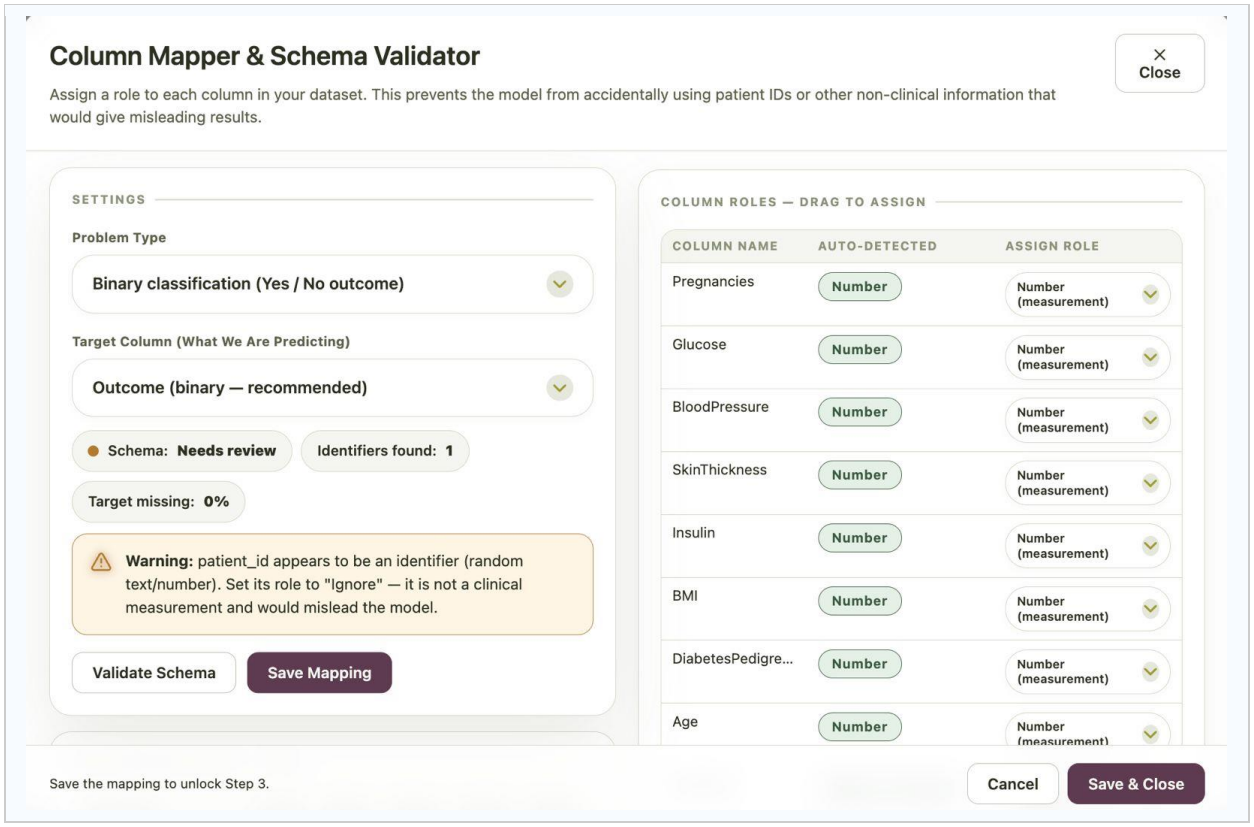
Feature 2.3 – Column Summary and Feature Overview

PATIENT MEASUREMENTS (FEATURES)

MEASUREMENT	TYPE	MISSING?	ACTION NEEDED
Pregnancies	Number	0%	Ready
Glucose	Number	0%	Ready
BloodPressure	Number	0%	Ready
SkinThickness	Number	0%	Ready
Insulin	Number	0%	Ready
BMI	Number	0%	Ready
DiabetesPedigreeFunction	Number	0%	Ready
Age	Number	0%	Ready
Outcome	Binary (2 values)	0%	Target (outcome)

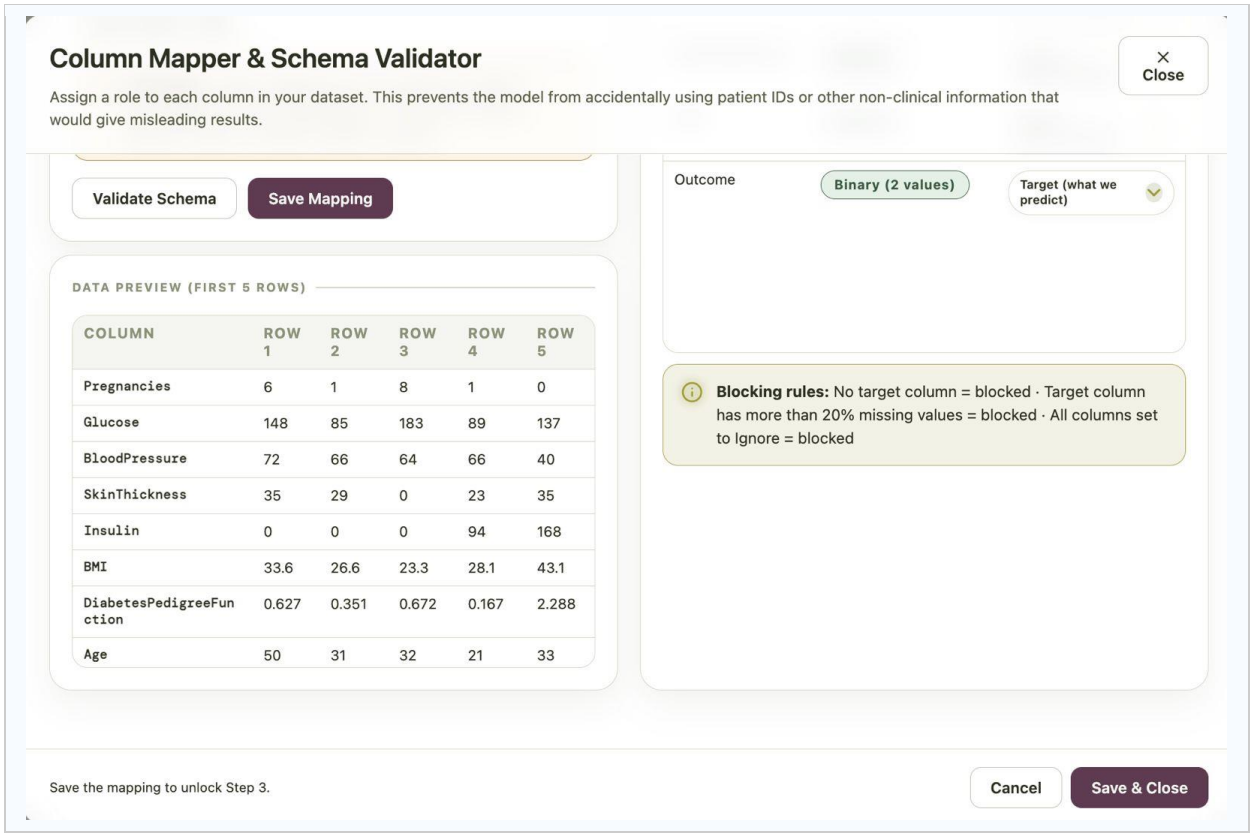
Explanation: This screenshot shows the column summary table generated during the Data Exploration step. The system automatically analyzes each dataset column and displays its data type, missing value percentage, and readiness status. The target variable (Outcome) is clearly identified, while the remaining columns are recognized as input features for the machine learning model.

Feature 2.4 - Column Mapper and Schema Validation (Before Validation)



Explanation: This screenshot shows the Column Mapper interface before schema validation. Users can assign roles to each dataset column (e.g., measurement, target, or ignore) to ensure that only relevant clinical features are used by the model. The system also detects potential identifiers such as patient IDs and warns the user to exclude them, helping prevent data leakage and misleading model predictions.

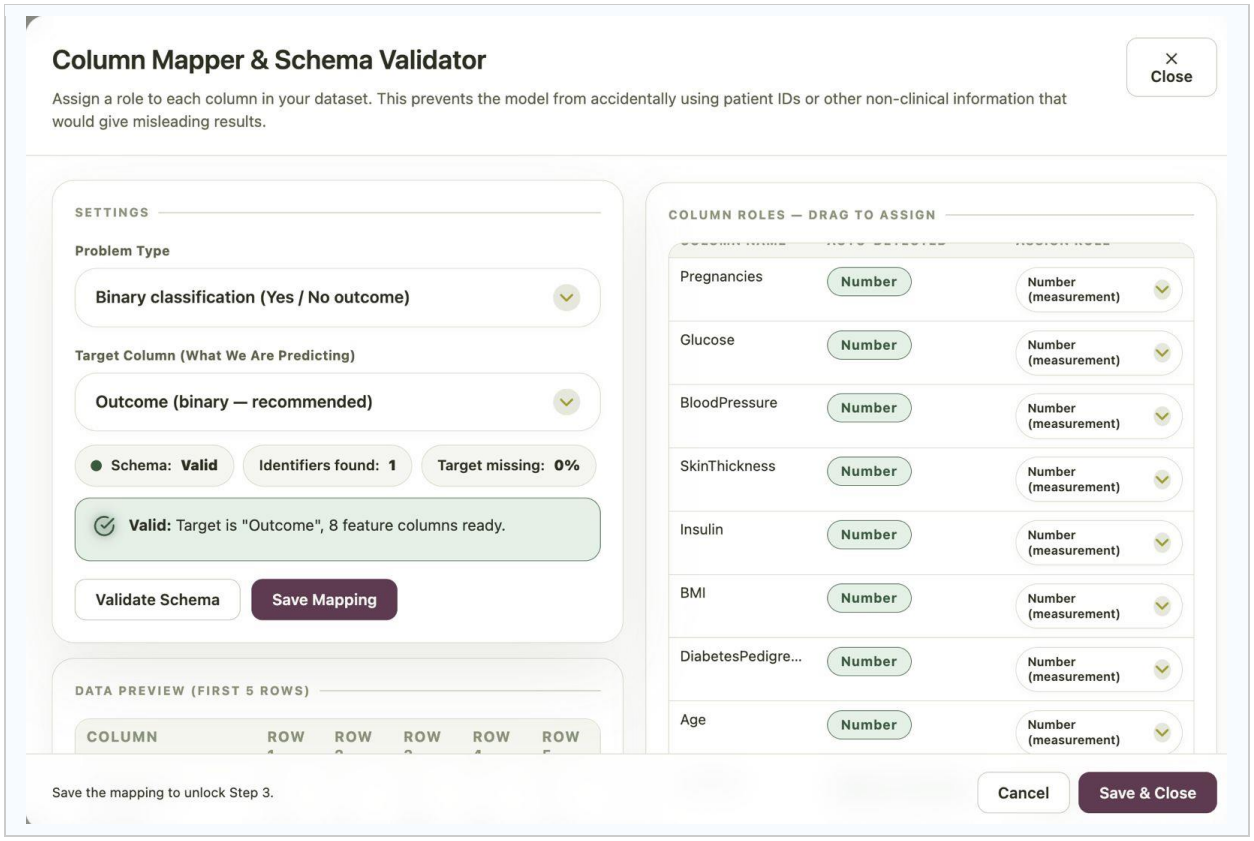
Feature 2.5 - Dataset Preview in Column Mapper



COLUMN	ROW 1	ROW 2	ROW 3	ROW 4	ROW 5
Pregnancies	6	1	8	1	0
Glucose	148	85	183	89	137
BloodPressure	72	66	64	66	40
SkinThickness	35	29	0	23	35
Insulin	0	0	0	94	168
BMI	33.6	26.6	23.3	28.1	43.1
DiabetesPedigreeFunction	0.627	0.351	0.672	0.167	2.288
Age	50	31	32	21	33

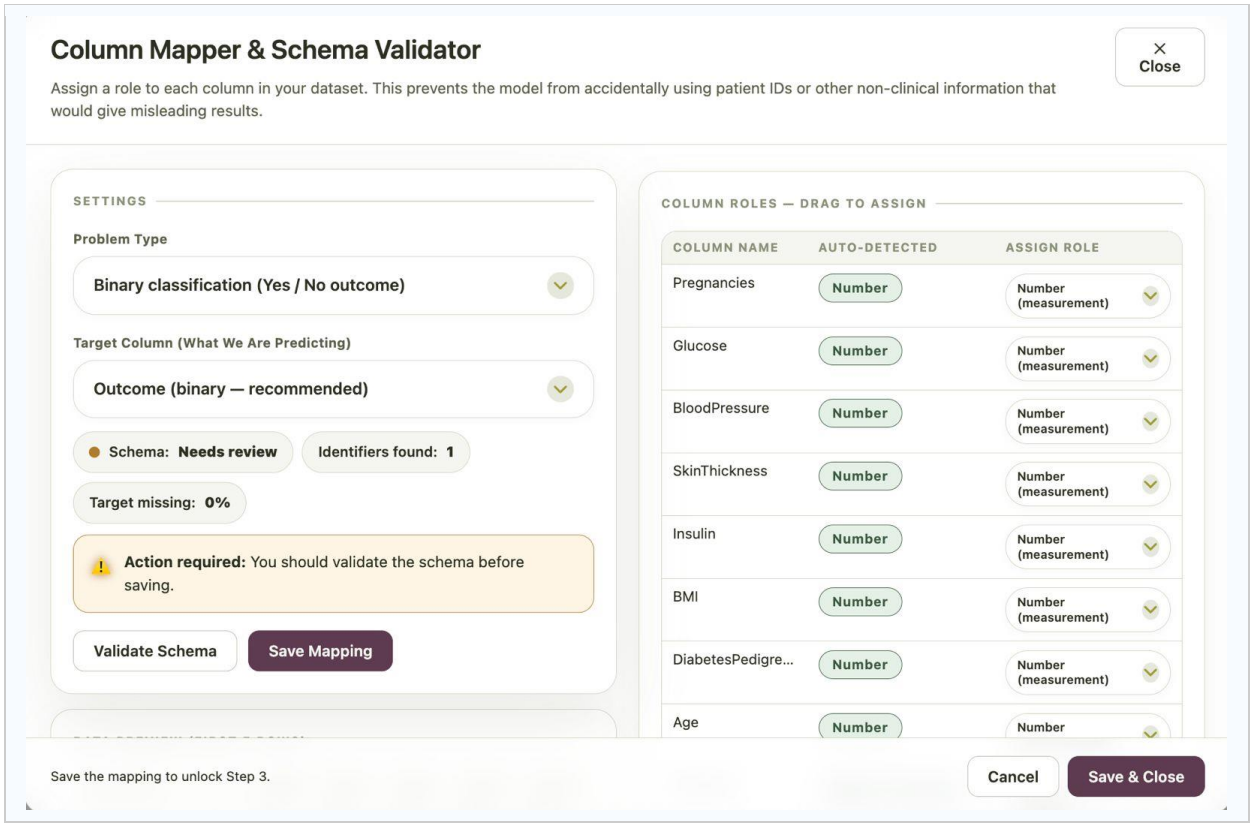
Explanation: This screenshot shows the dataset preview available within the Column Mapper interface. The tool displays the first rows of the dataset so that users can visually inspect the data before validating the schema. This helps confirm that columns are correctly interpreted and that the selected target and feature columns correspond to the actual dataset values.

Feature 2.6 - Schema Validation Success and Column Mapping Saved



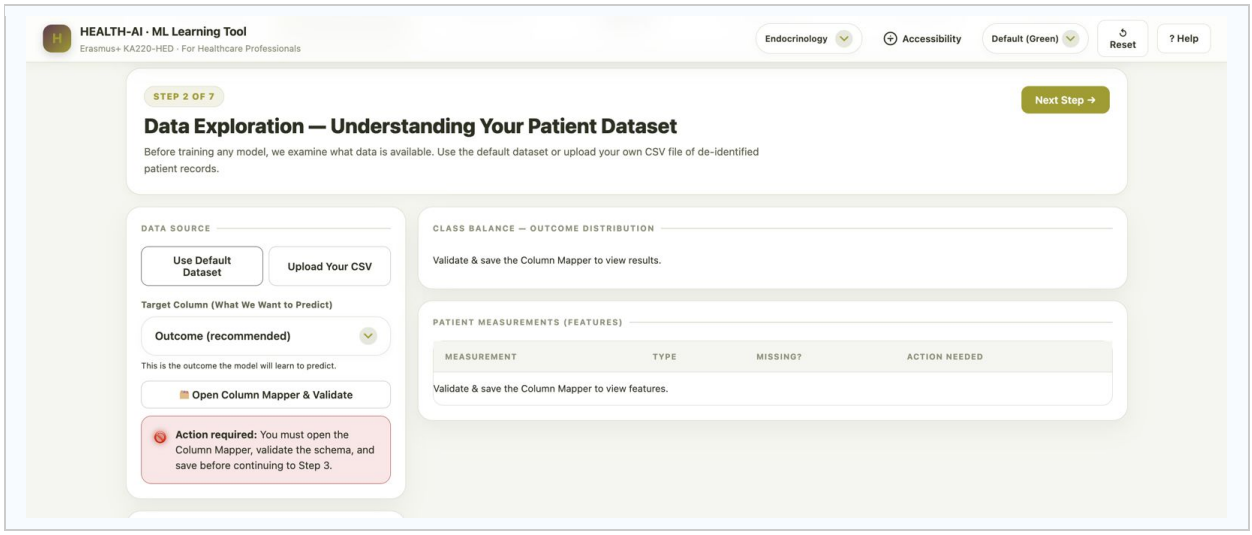
Explanation:
This screenshot shows the Column Mapper after successful schema validation. The system confirms that the target column ("Outcome") is correctly defined and that all feature columns are properly assigned. Once the mapping is saved, the dataset schema is validated and the workflow is unlocked, allowing the user to proceed to Step 3 (Data Preparation).

Feature 2.7 - Schema Validation Required Before Saving Mapping



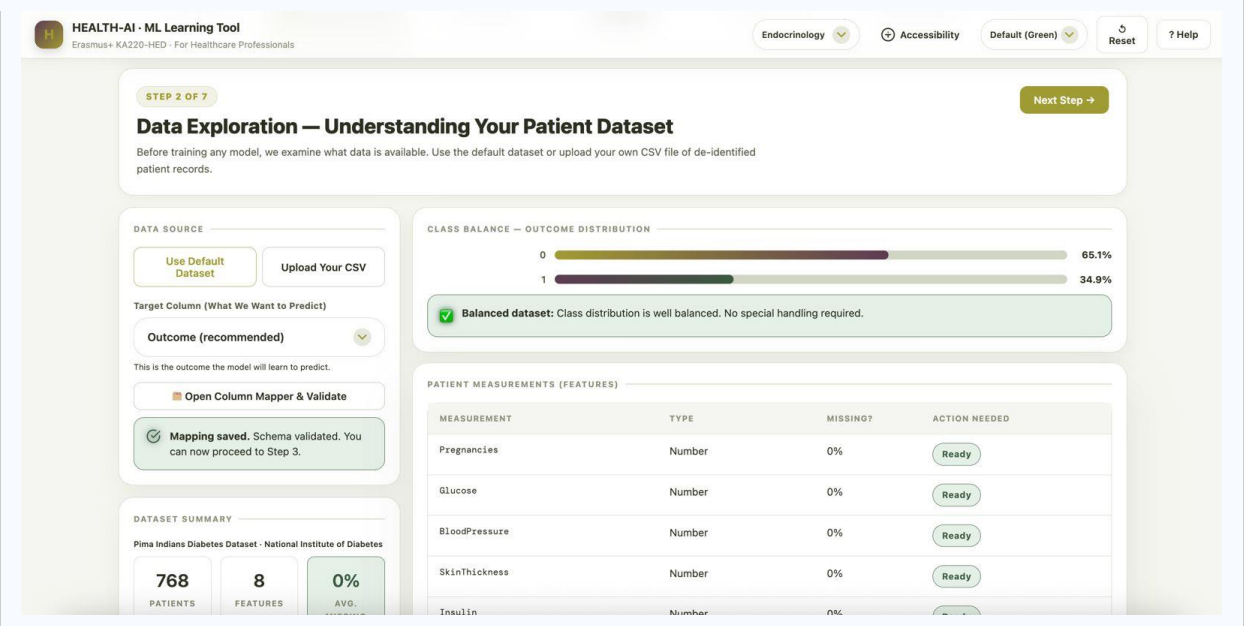
Explanation: This screenshot shows the system preventing the user from saving the column mapping before validating the dataset schema. When the user attempts to proceed without running schema validation, the tool displays a warning message indicating that validation is required. This ensures that the dataset structure is verified before the workflow continues, preventing incorrect configurations from entering the machine learning pipeline.

Feature 2.8 - Workflow Gate Control Before Proceeding to Step 3



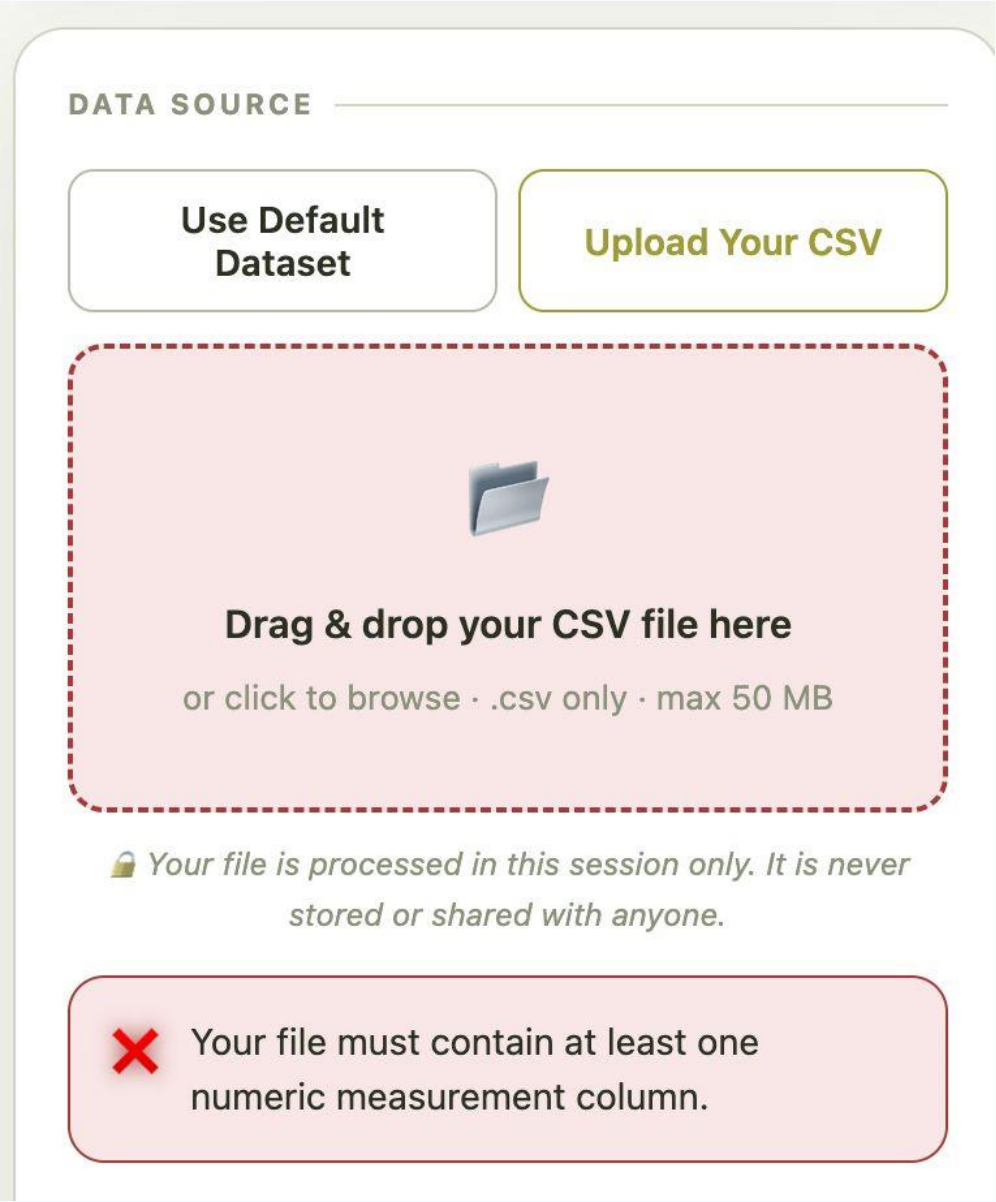
Explanation: This screenshot shows the workflow gate control mechanism in the Data Exploration step. When the user attempts to proceed to Step 3 without completing and saving the Column Mapper schema validation, the system blocks the action and displays a warning message. This ensures that the dataset schema is correctly validated before the machine learning pipeline continues to the Data Preparation stage.

Feature 2.9 - Workflow Unlocked After Successful Schema Validation



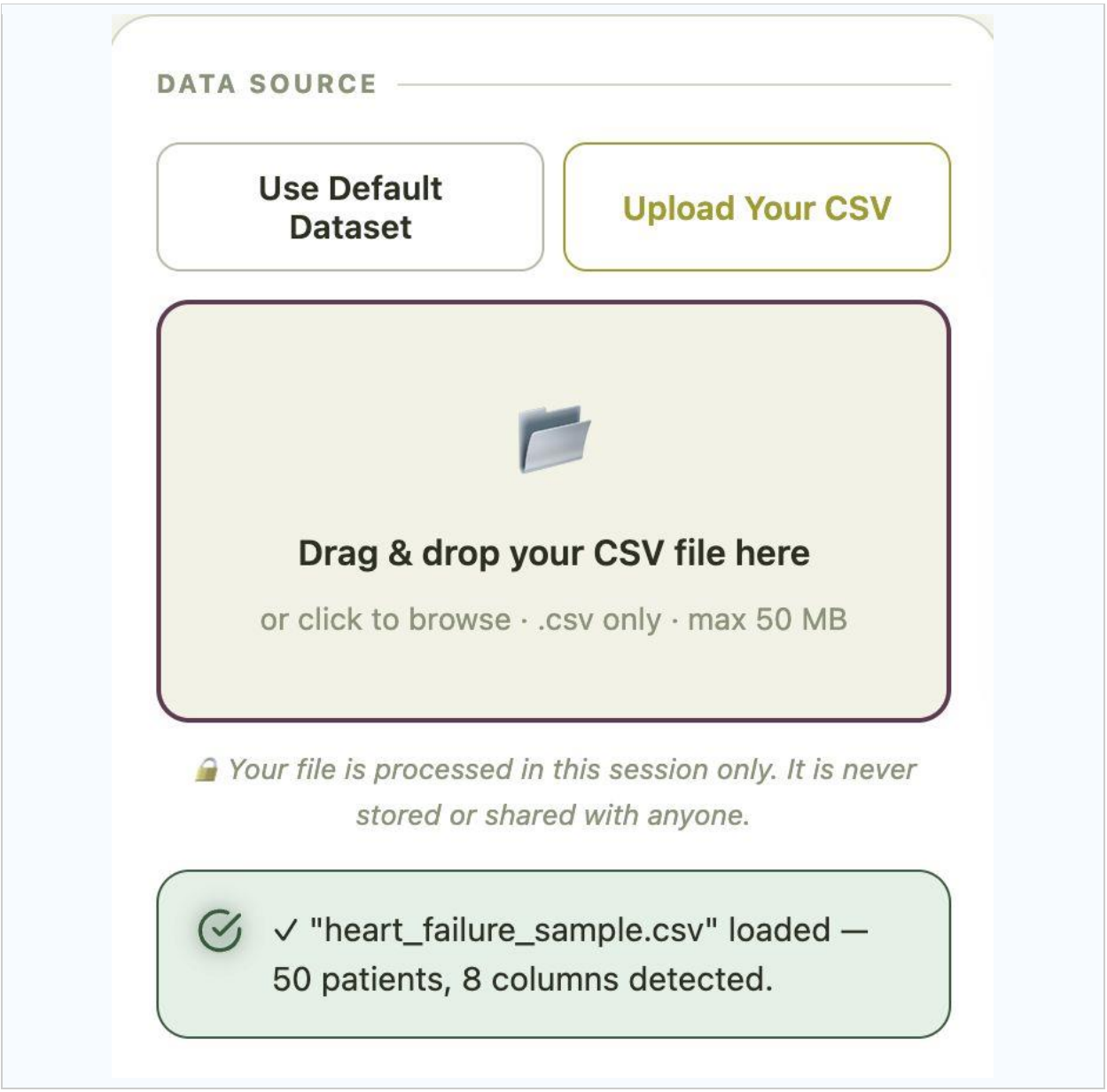
Explanation: [This screenshot shows the Data Exploration interface after the column mapping has been successfully validated and saved. The system confirms that the schema is valid and indicates that the user can now proceed to Step 3 (Data Preparation). This demonstrates the completion of the schema validation process and the unlocking of the next stage in the machine learning workflow.

Feature 2.10 - CSV File Upload with Validation and Error Handling



Explanation: This screenshot demonstrates the custom CSV upload functionality in the Data Exploration step. Users can upload their own dataset by dragging and dropping a CSV file. The system automatically validates the file structure and displays a clear error message when the dataset does not meet the required conditions (e.g., missing numeric measurement columns), ensuring that only valid datasets enter the machine learning workflow.

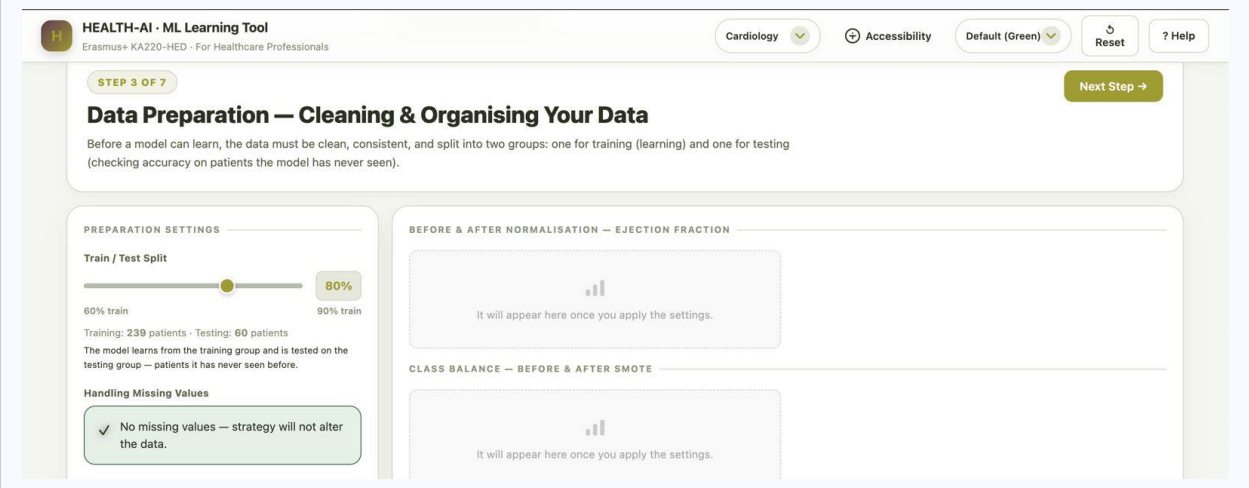
Feature 2.11 - Successful CSV File Upload and Dataset Detection



Explanation: This screenshot shows a successful custom dataset upload in the Data Exploration step. After uploading a valid CSV file, the system processes the dataset and displays a confirmation message indicating the number of patients (rows) and detected columns. This feedback confirms that the dataset has been correctly loaded and is ready for further exploration and schema validation.

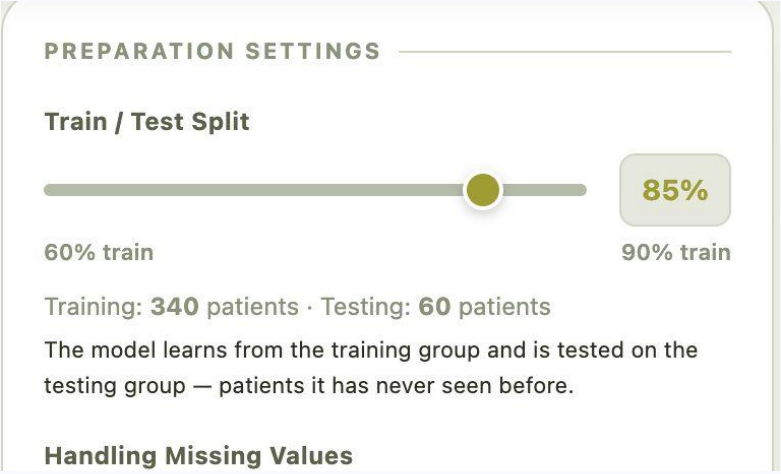
Step 3 – *Data Preparation*

Feature 3.1 – Training and Testing Data Split Configuration



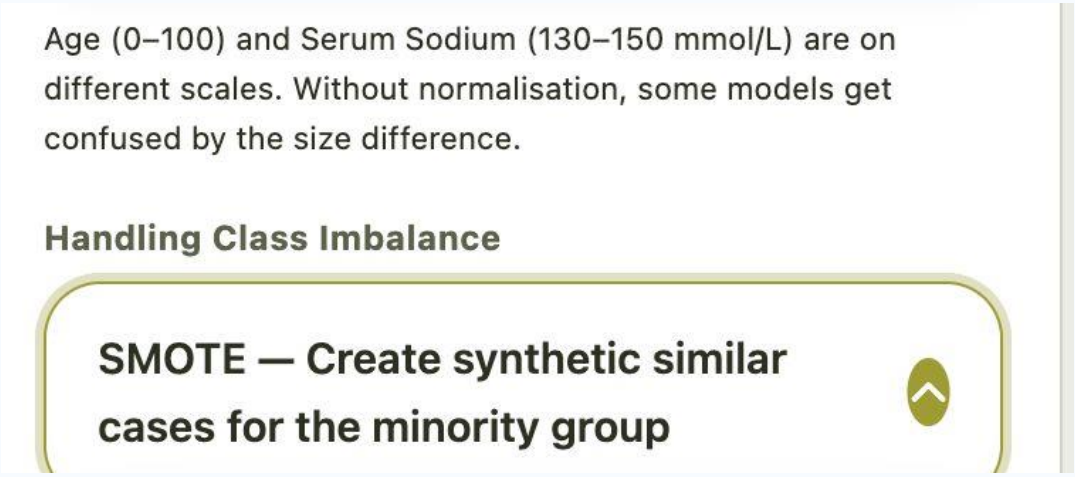
Explanation: This screenshot shows the training and testing split configuration in the Data Preparation step. Users can adjust the proportion of data used for model training and testing through an interactive slider. The interface dynamically updates the number of patient records allocated to each group, helping users understand how machine learning models are trained and evaluated on unseen data.

Feature 3.2 – Missing Value Handling Strategy Selection



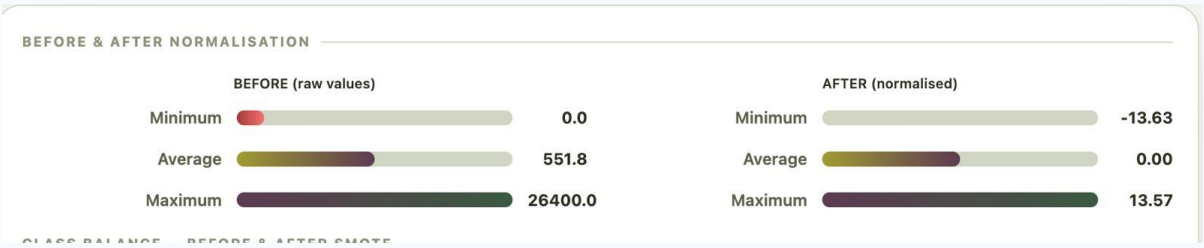
Explanation: This screenshot shows the configuration of missing value handling during the Data Preparation step. Users can select how missing values in the dataset should be handled before training the model. Available strategies include filling missing values with the median (recommended for numerical medical data), filling with the most frequent value (mode), or removing records that contain missing values. This allows users to understand and control how incomplete medical data is processed before machine learning training.

Feature 3.3 – Class Imbalance Handling Configuration



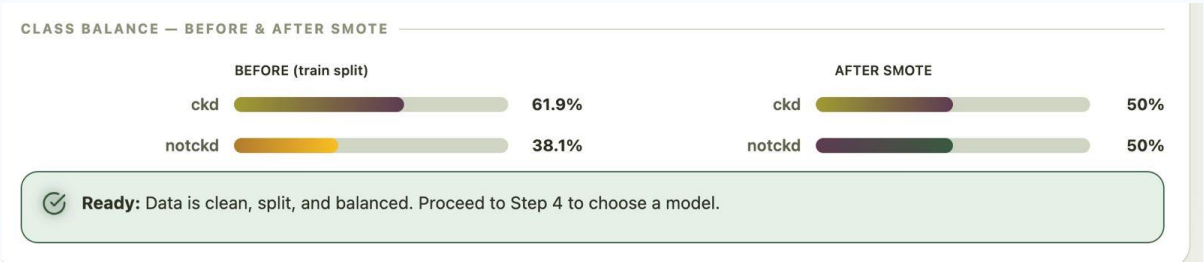
Explanation: This screenshot shows the configuration for handling class imbalance during the Data Preparation step. When the dataset contains significantly fewer samples in one class (minority class), the system offers several strategies to improve model learning. Users can select SMOTE (Synthetic Minority Over-sampling Technique) to generate synthetic samples for the minority class, apply class weighting to give higher importance to underrepresented cases, or proceed without applying any balancing technique.

Feature 3.4 - Data Normalisation Visualisation (Before and After)



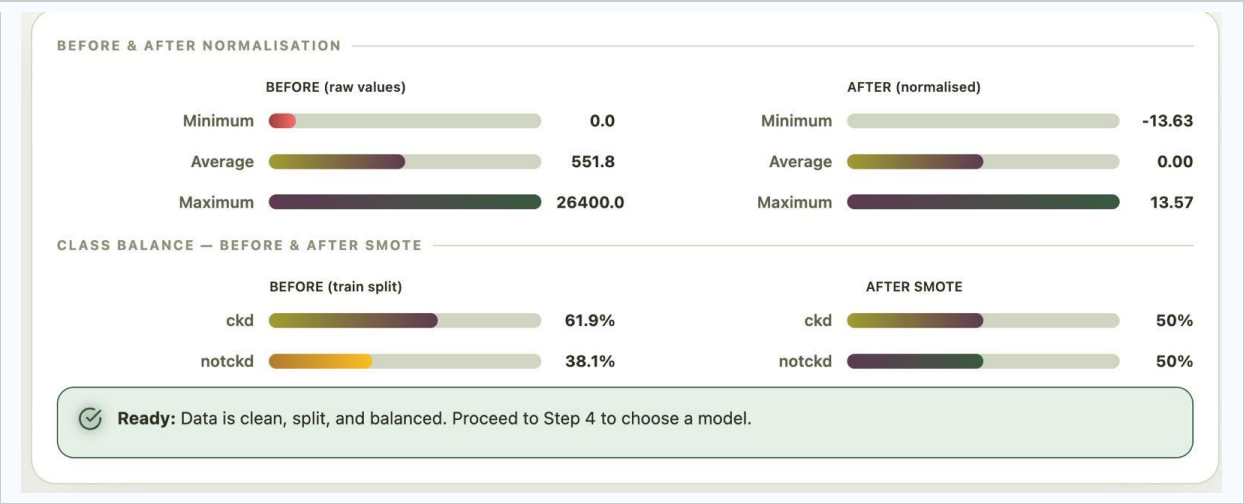
Explanation: This screenshot illustrates the effect of feature normalisation during the Data Preparation step. The interface compares the statistical distribution of the dataset before and after normalisation, showing the minimum, average, and maximum values. Normalisation places all numerical features on a comparable scale, which helps many machine learning models perform more effectively and prevents large-value features from dominating the learning process.

Feature 3.5- Class Balance Visualisation Before and After SMOTE



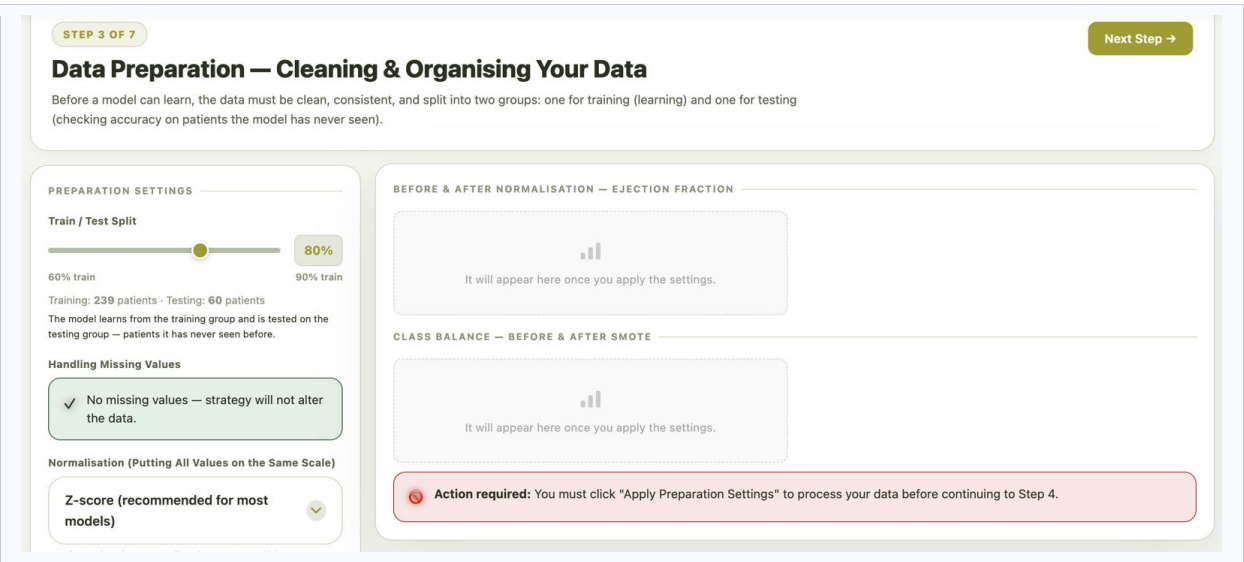
Explanation: This screenshot demonstrates the effect of applying SMOTE (Synthetic Minority Over-sampling Technique) during the Data Preparation step. The interface compares the class distribution before and after SMOTE is applied. Initially, the dataset shows an imbalance between the CKD and non-CKD classes. After applying SMOTE, synthetic samples are generated for the minority class, resulting in a balanced class distribution that helps machine learning models learn more effectively.

Feature 3.6- Data Preparation Completion Confirmation



Explanation: This screenshot shows the final confirmation message after all data preparation steps have been applied. The system indicates that the dataset has been successfully cleaned, split into training and testing sets, normalised, and balanced. Once these preprocessing steps are completed, the tool confirms that the dataset is ready and the user can proceed to the next step of the machine learning workflow, which involves selecting and training a predictive model.

Feature 3.7- Pipeline Gate Control Before Applying Preparation Settings



Explanation: This screenshot demonstrates the workflow control mechanism in the Data Preparation step. If the user attempts to proceed to the next stage without applying the selected

preprocessing settings, the system blocks the transition and displays a clear warning message. This ensures that data cleaning, splitting, normalisation, and balancing operations are properly executed before the machine learning model training step begins.